

Hernán Díaz Rodríguez
Department of Computing, University of Oviedo
Gijón, Spain
diazhernan@uniovi.es

9 October 2026

To the Editor-in-Chief,
Applied Soft Computing

I submit for your consideration the manuscript *Genetic Programming Hyper-Heuristics for the Job Shop Scheduling Problem with Interval Durations: Robustness-Aware Interpretable Rules that Generalize Across Instance Sizes*, as a technical paper.

In many manufacturing settings the duration of an operation is not known in advance, and a planner can only give a lower and an upper bound. The interval job shop scheduling problem models this imprecision without assuming any probability distribution, and its best solutions so far come from metaheuristics that search anew for every instance. The manuscript brings genetic programming to this problem. A hyper-heuristic evolves dispatching rules that read both the worst-case value and the width of each uncertain quantity. The rules build a schedule in milliseconds and are short enough to be read and checked by a planner.

The contribution has three parts. First, rules evolved on four instances improve on every constructive baseline on all seventy instances of the benchmark, and transfer without retraining to all size classes and to a second benchmark. Second, ablations show where interval information pays off: not in the expected makespan, but in the predictability of the schedule. An objective that penalizes the width of the predicted makespan yields schedules whose execution departs less from the prediction, with a single weight that sets the trade-off. Third, a comparison at matched budget with a genetic algorithm, reimplemented on the same decoder, shows up to which computational budget the evolved rule is the better choice, and how that budget grows with instance size.

The work fits the scope of the journal on two counts. It applies evolutionary computing to a real-world complexity, uncertain processing times. And it does so in the spirit the journal describes: it exploits the tolerance for imprecision that intervals express to obtain schedules at low solution cost, and examines what robustness that tolerance can buy. The manuscript also builds on genetic programming hyper-heuristics for scheduling published in the journal.

I confirm that the manuscript is original, that it has not been published elsewhere, and that it is not under consideration by another journal. An earlier version of this study is available as a preprint (SSRN, doi:10.2139/ssrn.7214286).

The complete source code is openly available in Zenodo (doi:10.5281/zenodo.21716972; version 2.0, which accompanies this manuscript: doi:10.5281/zenodo.23156759): the hyper-heuristic, the baselines and the genetic algorithm, together with a test that re-derives the deposited results from scratch. The deposit also holds the instances, the evolved rules and the result files behind every table and figure. A statement on the use of generative AI in the preparation of the manuscript appears directly before the references, as the journal's policy requires.

Thank you for considering this work.

Yours sincerely,

Hernán Díaz Rodríguez
Corresponding author